

Nowcasting Real GDP Growth in The Bahamas

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Motivation: The data timeliness problem...

- The Bahamas 1Q2025 GDP released **5 months** after the standard (e. g., May)
- Globally: Only 55% of countries publish quarterly GDP timely (Berry et al., 2018)
- Critical for small, open economies vulnerable to:
 - Natural disasters
 - Other external shocks
- Also important for timely decision-making
- **Solution:** Nowcasting with high-frequency indicators

Motivation: ...coupled with access challenges

TABLE 2: GROSS DOMESTIC PRODUCT BY ECONOMIC ACTIVITY (B\$ MILLIONS)

REAL I Quarter 2016 - II Quarter 2024. (2018 = 100)

Industry Classification (Revision 4)	ISIC Code	Q1 2023 R	Q2 2023 R	Q3 2023 R	Q4 2023 R	Q1 2024 P	Q2 2024 P
Agriculture, forestry and fishing	A	16.0	13.1	15.3	23.1	23.3	19.8
Mining	B	24.9	26.1	27.4	23.2	32.5	28.1
Manufacturing	C	12.4	11.4	19.4	22.2	11.8	20.1
Electricity and gas, Water supply and sewerage	D & E	59.8	57.8	62.2	46.6	48.0	51.8
Construction	F	160.8	155.8	169.3	147.7	131.6	241.8
Wholesale and retail trade, Motor Vehicle repairs	G	331.2	325.3	304.1	305.6	306.9	309.7
Transport and Storage	H	152.9	161.0	198.3	180.8	162.1	180.2
Accommodation and Food Services	I	289.1	287.0	286.5	285.9	285.0	313.9
Information and Communication	J	115.4	100.4	121.2	120.3	128.2	129.6
Financial and Insurance activities	K	261.9	253.4	232.8	235.1	231.3	240.4
Real Estate activities	L	529.6	520.0	508.2	507.1	536.8	500.5
Professional, scientific and technical services	M	125.8	129.5	114.0	123.8	123.7	128.4
Administrative and support services	N	84.3	84.0	78.4	79.9	83.4	82.8
Public administration and defense, Social security	O	255.1	257.3	255.9	254.6	253.4	252.3
Education	P	79.1	79.6	79.5	79.5	79.4	79.3
Human health and social work	Q	96.8	99.1	99.7	99.8	99.5	99.2
Arts, Other Services, Household Employment, Extraterrestrial Org.	R, S&T	289.0	321.3	291.9	299.5	323.6	308.7
Total GDP at Basic Prices		3,864.2	3,882.0	3,864.2	3,814.7	3,860.8	3,974.9
Taxes less subsidies on products		258.5	207.2	256.1	186.5	223.1	262.8
Sub Total		3,122.7	3,089.2	3,120.2	3,001.2	3,088.8	3,237.7
Statistical discrepancy		125.9	124.7	128.1	121.3	124.7	130.9
GDP by economic activity at purchaser current prices		3,248.6	3,213.9	3,246.3	3,122.5	3,208.5	3,368.7
Growth Rate (PREVIOUS QTR)		3.8%	-1.3%	1.0%	-3.8%	2.8%	5.0%
COMPARED TO THE SAME QUARTER IN PREVIOUS YEAR		0.52%	3.58%	1.55%	-0.61%	-1.23%	4.61%

(a) BNSI's National Accounts Table

• Complex PDF formats

• Scattered publications

• Manual processing required

Table 8.18 Selected Economic Indicators

INDICATOR	Unit	2017	2018	2019	2020	2021	2022	2023	2024
Nonoil Exports (f.o.b.)	B\$000	400,627	411,821	457,867	276,399	358,851	424,831	687,250	663,487
Nonoil Imports (c.i.f.)	B\$000	2,874,958	2,938,016	2,551,720	1,818,573	2,824,819	3,153,452	3,562,393	4,139,121
Average Retail Price Index	Nov. 2014=100	103.20	105.54	108.17	108.21	111.35	117.59	121.19	121.68
Total Tourist Arrivals	(000)	6,136	6,622	7,250	1,795	2,101	7,001	9,655	11,217
Value of Construction Permits*	B\$000	423,408	664,626	705,873	686,715	706,469	988,762	915,565	541,498
Value of Construction Starts*	B\$000	136,615	118,174	102,881	154,564	313,714	340,161	212,104	169,738
Value of Construction Completions*	B\$000	1,483,807	333,784	212,978	200,051	307,293	435,453	391,530	245,054
Government Revenue (Calendar Year) ¹	B\$000	2,087,215	2,173,343	2,503,238	1,666,737	2,369,160	2,730,278	2,899,257	3,208,007
Government Revenue (Fiscal Year: Jul-Jun) ²	B\$000	2,070,259	2,042,385	2,426,318	2,082,003	1,908,776	2,605,701	2,855,445	3,069,106
Government Expenditure (Calendar Year) ³	B\$000	2,709,709	2,510,709	2,754,918	3,032,853	3,250,939	3,448,513	3,414,734	3,547,384
Government Expenditure (Fiscal Year: Jul-Jun) ⁴	B\$000	2,730,986	2,457,286	2,645,584	2,920,514	3,243,583	3,327,394	3,390,028	3,263,116
Government Debt (Direct Charge) ⁵	B\$000	7,180,089	7,498,912	7,733,214	9,417,448	10,317,372	11,035,946	11,427,480	11,767,607
Average Treasury Bill Discount Rate	%	1.89	1.71	1.75	1.93	2.85	2.88	2.91	2.94
Money Supply (M1)	B\$000	2,654,036	2,728,160	3,248,398	3,472,120	3,715,501	4,296,761	4,318,660	4,626,451
Money Supply (M2)	B\$000	6,763,155	6,707,279	7,304,977	7,505,689	7,772,682	8,459,695	8,602,888	8,989,337
Money Supply (M3)	B\$000	7,037,296	7,108,822	7,892,847	7,864,180	8,220,340	9,002,044	9,133,858	9,485,830
Bank Credit (all currencies)	B\$000	8,836,328	8,911,192	8,957,100	8,614,408	8,928,992	9,079,562	9,376,312	9,871,098
Bank Deposits (all currencies)	B\$000	6,925,956	6,913,198	7,727,494	7,731,526	7,976,839	8,716,755	8,911,633	9,181,146

SOURCE: Data compiled from various tables in the Digest.
NOTE: * Excludes Family Islands' Statistics.

(b) CBOB's Quarterly Statistical Digest Table

Why Nowcasting?

- **Nowcasting:** Particularly valuable when (Akbal et al., 2023):
 - Official data has long publication lags
 - High-frequency indicators are available
 - Elevated uncertainty
 - Vulnerability to external shocks
- **Our contribution:** First *comprehensive* nowcasting framework for a Caribbean economy

Comprehensive Dataset Structure (359 variables total)

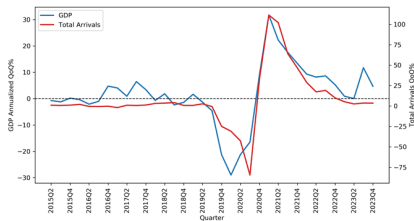
Structured Data (218 series)

- Tourist arrivals
- Fiscal accounts
- Trade flows
- Monetary and financial indicators

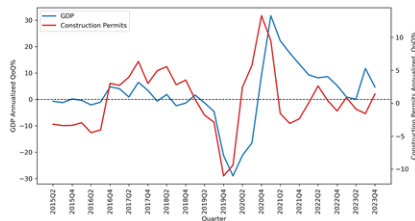
Unstructured Data (141 series)

- Google Trends (101 variables)
- Satellite NTL (40 variables)

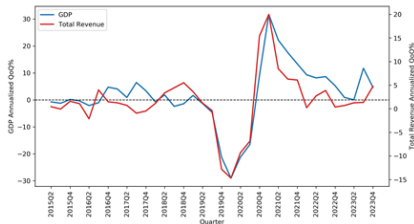
Structured data: Key Economic Indicators vs GDP



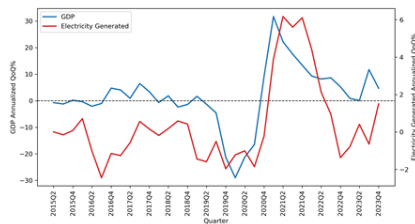
(a) Tourist Arrivals



(b) Construction Activity



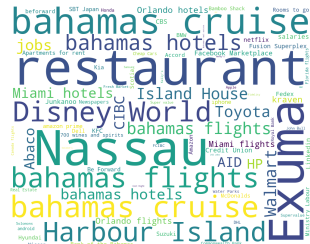
(c) Government Revenue



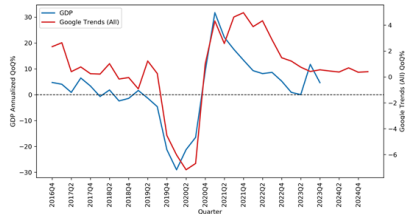
(d) Electricity Generation

Unstructured data: Google Trends

- **101 search variables** retrieved from a survey in Bahamas COF tracking demand for private consumption
- Categories include:
 - Tourism: "Bahamas flights", "Bahamas hotels", "Bahamas cruise"
 - Retail: "Walmart", "Toyota", "Facebook Marketplace"
 - Labor market: "LinkedIn", "242 Jobs", "FedEx"
- **Frequency:** Real-time data with 1-week lag
- Powerful for tracking tourism and consumption in real-time (Choi and Varian, 2012; Kohns and Bhattacharjee, 2023)

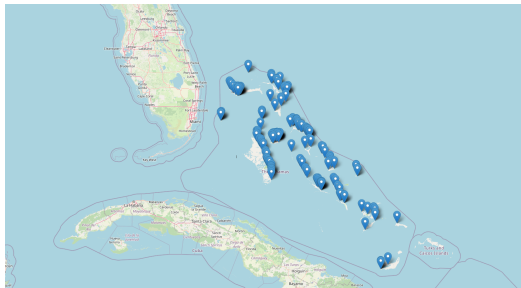


(a) Word cloud: correlation with GDP



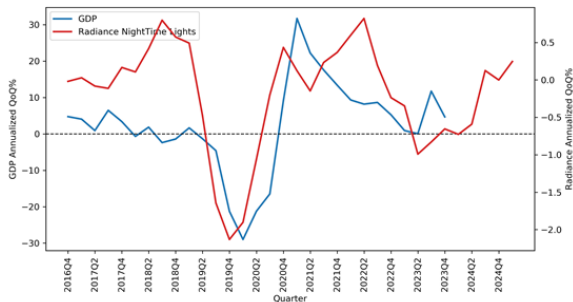
(b) Google Trends vs GDP

Unstructured data: Nighttime Lights



(a) NTL measurement locations

- 198 hotels from MoT database
- 7 airports, 16 ports, 9 consumption centers
- Daily VIIRS satellite data

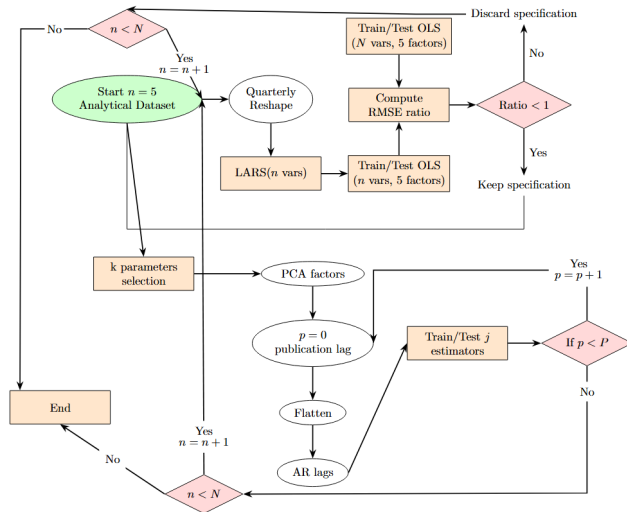


(b) NTL vs GDP growth

- Strong co-movement with GDP
- Good proxy of tourism activity (Henderson et al., 2012; Fontaine et al., 2025)

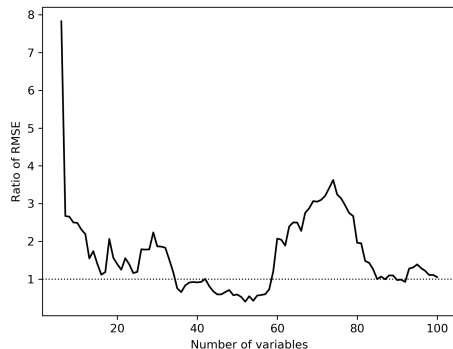
Methodology: Multi-Step Nowcasting Framework

- Variable pre-selection via LARS (28 optimal groupings)
- Multiple specifications (PCA factors, AR lags, flattening)
- 13 diverse estimators from econometrics and ML



Variable pre-selection: Least Angle Regression (LARS)

- **Purpose:** Handle high-dimensional data (359 variables)
- **Process:**
 - 1 Start with all coefficients at zero
 - 2 Incrementally add most correlated predictors
 - 3 Adjust coefficients in equiangular direction
 - 4 Stop at optimal number of variables
- **Result:** Identified 28 optimal variable groupings
- **Benchmark:** Outperforms OLS using first five principal components of full dataset



LARS RMSE vs Benchmark

Superior to other selection methods (Chinn et al., 2023)

Specifications: Key parameter variation

Parameters vary systematically:

- ① **Flattening lags** to handle mixed frequencies:
 - 1.1 3 lags (baseline) - 28 specifications
 - 1.2 4 lags for linear models and 5 lags for ML models (expanded) - 28 specifications
- ② **PCA factors** for dimensionality reduction:
 - 2.1 No factors - 56 specifications
 - 2.2 3 factors - 56 specifications
 - 2.3 5 factors - 56 specifications
- ③ **AR lags** to capture macroeconomic dynamics:
 - 3.1 No lags - 28 specifications
 - 3.2 2 lags - 28 specifications

Total: 197 specifications addressing core nowcasting challenges

Includes 1 human expert specification for benchmarking

Comprehensive model collection: 13 estimators

Econometric Models

- ARMA
- OLS
- MIDAS (Ghysels et al., 2016)
- DFM (Bok et al., 2018)
- BVAR (Ankargren and Jonéus, 2020)

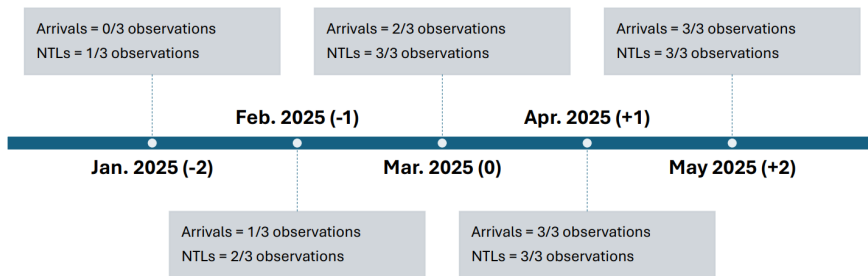
Regularization Models

- Ridge (Hoerl and Kennard, 1970)
- LASSO (Tibshirani, 1996)
- Elastic Net (Zou and Hastie, 2005)

Machine Learning Models

- Decision Trees (Breiman et al., 1984)
- Gradient Boosted Tree (Friedman, 2001)
- Random Forest (Breiman, 2001)
- LSTM (Hochreiter and Schmidhuber, 1997)
- MIDAS-ML (Babii et al., 2022)

Testing: Assessing accuracy through the nowcast cycle



- Five distinct time horizons (Chinn et al., 2023; Hopp, 2024)
- From 2 months before to 2 months after quarter end
- Accounts for asynchronous data publication
- Tracks nowcast improvement with information flow

Technical challenges addressed

Curse of Dimensionality

- LARS variable pre-selection (Bai and Ng, 2008; Chinn et al., 2023)
- PCA (Stock and Watson, 2002; Chinn et al., 2023)

Macroeconomic Dynamics

- AR lags of GDP (Jansen et al., 2016)

Mixed Frequencies & Ragged Edges

- Vertical realignment (Hopp, 2024)

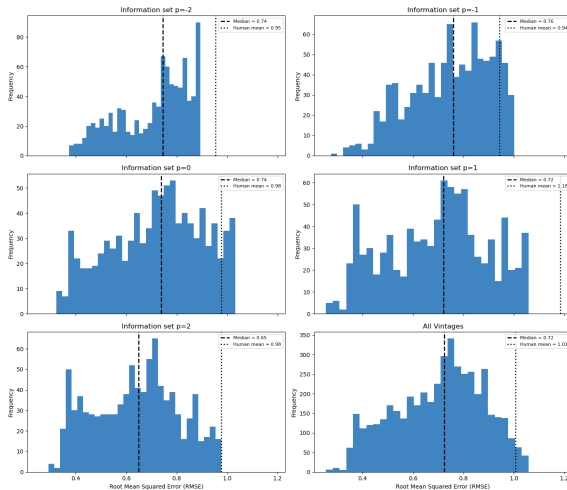
Model Uncertainty

- Econometrics + ML (Hopp, 2024; Kant et al., 2025)
- Model weighting (Kitchen and Monaco, 2003)
- **Distribution of nowcasts**

Key innovation: ensemble distribution approach

- **Traditional approach:** Single “best” model specification
- **Our approach:** Distribution of nowcasts from multiple models
- **Process:**
 - ① Generate 1,858 estimates (groupings $\times$ specifications $\times$ estimators)
 - ② Filter: Keep only models with RMSE below median
 - ③ Result: Distribution with confidence intervals
 - ④ Weight: Inverse RMSE weighting for final ensemble
- **Advantage:** Captures uncertainty, leverages diverse model strengths

Results: Nowcast Distribution - RMSE



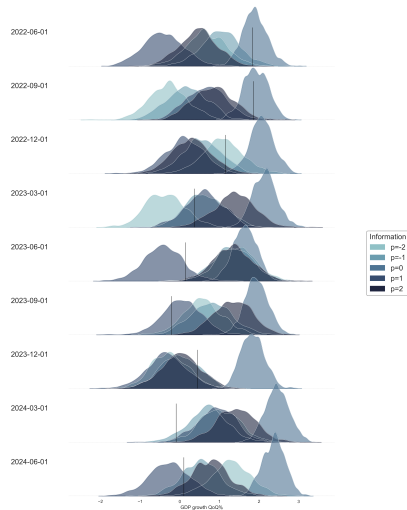
Results: ND improves performance

Model	RMSE	Improvement
Human Benchmark	1.01 pp	–
Ensemble Nowcast	0.72 pp	29%

- Median error below 0.80 pp
- Performance improves with information availability
- Most specifications beat human benchmark

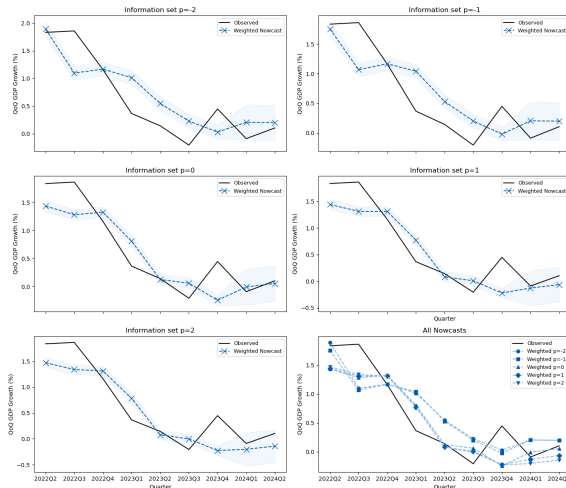
Results: ND across the nowcast cycle

- **Distributions of nowcasts** for best-performing models across vintages and test periods
- Vertical lines indicate observed GDP growth
- Distributions contribute to addressing uncertainty



Results: ND ensemble improves accuracy over time

- Successfully captures GDP growth direction and, to a good extent magnitude
- Accuracy improves as more data becomes available



Concluding remarks

Core Innovation

Distribution of nowcasts rather than point estimates

Key Contributions

- First comprehensive nowcasting framework for a Caribbean economy
- Integration of unconventional data

Policy Impact

- 2–8 month lead on official GDP releases
- Real-time economic monitoring
- Scalable framework for data-constrained environments

Performance Results

- **29% improvement on average** over expert benchmark
- Median RMSE consistently below 0.80 pp
- Accurate direction and magnitude in out-of-sample tests
- Robust over the nowcasting cycle

Q&A

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